

Performance QA Engineer: Test Assignment

Your task is to conduct performance testing for [YouTrack](#). The goal is to assess whether the application can handle the target load and determine how close the system operates to its limits under that load.

Test Environment Setup

You should deploy and test the Server version of the YouTrack application.

- Installation instructions can be found here:
https://www.jetbrains.com/youtrack/download/get_youtrack.html?section=server
- Use version **2026.1.13456**
- A 60-day trial license for 100 users is provided (the license is attached in a separate file)
- Recommended resources for YouTrack are at least:
 - 2 CPU cores
 - 3 GB RAM
- Use the default application settings
- The load generator may run either on the same machine as the tested application or on a separate machine, depending on your preference and available resources

If running the application locally is inconvenient for you, please contact us and we will consider providing a cloud-based YouTrack instance.

Test Data Preparation

Before running the tests, you should populate your YouTrack instance with test data consisting of:

- 1 project
- 100 users
- 100,000 issues

Requirements:

- Inject the data into an empty tracker via the REST API
- You may use the default **Demo** project automatically created during YouTrack installation

- All users may be added to the Demo project team with default permissions, in the same way as on the following page:
<yourtrack_baseurl>/admin/editProject/Demo?tab=team
- All users could be part of the group Registered Users, and the group Registered Users could be able to view and modify the Demo project

When creating issues, ensure sufficient data variety:

- Each issue must have a summary
- Each issue must have a description
- Issues should be created on behalf of different users
- Values of other custom fields may remain default

For authorization, it is strongly recommended to use a permanent token instead of the standard login process.

Once the test data is injected, you may create a database backup so it can be restored before rerunning performance tests if needed.

Test Scenario

You should develop a test scenario that mimics end-user behavior.

Assume that during one hour, an average user:

- Creates or updates 10 issues
 (Updates may include commenting, changing fields, applying commands, adding links – your choice)
- Periodically opens other issues for reading
- Performs searches
 (Approximately 10 searches per hour and 30 viewed issues; issue selection is up to you. A fixed set of 50 or fewer distinct search queries shared across all users is sufficient.)

Test Requirements

Your goal is to design and execute performance tests in order to answer the following questions:

1. Can the tracker withstand a load of 100 concurrent active users under the given resource constraints?
2. What performance headroom exists – how many concurrent users can the system handle before performance degradation begins?

3. Is it possible to identify the application's breaking point under the selected resource constraints, or does something prevent this?

Assume that a violation of at least one of the following SLOs indicates performance degradation:

- 90th percentile response time for `GET /api/sortedIssues` ≤ 500 ms
- 90th percentile response time for `GET /api/issue/{id}?fields=...` ≤ 500 ms
- 90th percentile response time for `POST /api/issuesGetter?fields=...` ≤ 1000 ms
- HTTP 500 error rate $\leq 1\%$

You should avoid excessive test reruns. If you fail to reach the breaking point after the second or third attempt, you may either:

- Arbitrarily tighten the degradation criteria and draw conclusions based on that, or
- Explain why the proposed degradation point was not reached

One or two successful 60-minute runs are sufficient, although shorter tests are also acceptable.

Final Deliverable: Performance Testing Report

Provide a report in English covering:

- Test methodology description
- Summary of test results
- Statistics and graphs illustrating your findings
- A link to the repository containing the implemented test suite
- Any additional information or attachments that you believe should be included in the report

Technical Requirements

- **Gatling** is the preferred load testing tool for our project, but it is not a strict requirement – you may use any tool you prefer
- If you use AI agents to complete the assignment, please save `AGENTS.md`, `CLAUDE.md`, `skills/*.md`, or similar files in the repository together with the test suite
- You are free to use any tools and technologies you are comfortable with for any activities that, in your opinion, are required to complete this assignment.

Links to the YouTrack documentation

- System Requirements, Installation, and Configuration
 - <https://www.jetbrains.com/help/youtrack/server/youtrack-supported-environments.html>
 - <https://www.jetbrains.com/help/youtrack/server/youtrack-docker-installation.html>
- Database Backup
 - <https://www.jetbrains.com/help/youtrack/server/back-up-the-database.html>
 - <https://www.jetbrains.com/help/youtrack/server/restore-docker-image-installation.html#restore-from-database-backup>
- Issues (fields, comments, history)
 - <https://www.jetbrains.com/help/youtrack/server/reporting-an-issue.html>
- Issue Links
 - <https://www.jetbrains.com/help/youtrack/server/link-issues.html>
- Search and Search Syntax
 - <https://www.jetbrains.com/help/youtrack/server/attribute-based-search.html#attribute-based-search-syntax>
- Users, Groups, Permissions
(default project teams should be sufficient)
 - <https://www.jetbrains.com/help/youtrack/server/manage-user-groups.html#teams>
- Permanent Tokens
 - <https://www.jetbrains.com/help/youtrack/devportal/authentication-with-permanent-token.html>
 - <https://www.jetbrains.com/help/youtrack/devportal/Manage-Permanent-Token.html>
- Commands
(you can manage without them, but they greatly simplify field interactions, adding links, and even comments)
 - <https://www.jetbrains.com/help/youtrack/server/apply-commands-to-issues.html#apply-command-to-issue>
- REST API
(for this assignment, browser dev tools and/or a recording proxy from your preferred performance testing framework are sufficient, but you can read more here if needed)
 - <https://www.jetbrains.com/help/youtrack/devportal/youtrack-rest-api.html>